

Structure-based selectivity mapping of HIV-1 protease inhibitors against human aspartic-protease off-targets

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Independent computational research project · Molecular docking study · Data and code:
github.com/maxwo123/computational-biology

Abstract. Off-target inhibition of host enzymes is a major driver of drug toxicity and attrition in medicinal chemistry. HIV-1 protease inhibitors (PIs) act on a viral aspartic protease, but two human aspartic proteases — cathepsin D and β -secretase 1 (BACE1) — share the same catalytic architecture and are therefore plausible off-targets. Here we computationally map the selectivity of 20 HIV-1 PIs (spanning FDA-approved, discontinued, and investigational agents) by molecular docking against the viral target (HIV-1 protease, PDB 1HXW) and both human off-targets (cathepsin D, 1LYB; BACE1, 1FKN) using AutoDock Vina. A selectivity metric $\Delta\Delta G$ was defined as the difference between each compound's off-target and on-target predicted binding energy. The docking protocol was validated by native ligand re-docking, reproducing the crystallographic ritonavir pose in HIV-1 protease to 1.93 Å RMSD. 14 of 20 compounds were predicted to be HIV-selective (binding the viral protease more tightly than either human enzyme). Predicted selectivity correlated negatively with molecular polarity and size (topological polar surface area, $r = -0.59$; rotatable bonds, $r = -0.49$), suggesting that smaller, more rigid, less polar inhibitors preferentially engage the viral pocket. These results illustrate a reproducible, open-source workflow for early-stage off-target risk assessment.

1. Introduction

The protease encoded by human immunodeficiency virus type 1 (HIV-1) cleaves the viral Gag–Pol polyprotein into functional units and is essential for producing infectious virions. It is an aspartic protease that operates as an obligate homodimer, with a single catalytic site formed at the interface of two identical subunits and powered by a pair of catalytic aspartate residues (the "Asp dyad"). Inhibiting this enzyme halts viral maturation, and HIV-1 protease inhibitors (PIs) remain a cornerstone of antiretroviral therapy.

A central concern in drug design is *selectivity*: a therapeutic should engage its intended target while sparing structurally related host proteins. Human aspartic proteases such as cathepsin D (a lysosomal enzyme implicated in protein turnover and, when dysregulated, in cancer progression) and BACE1 (β -secretase, which generates the amyloid- β peptide central to Alzheimer's disease) use the same Asp-dyad catalytic chemistry as the viral enzyme. This shared active-site architecture creates a plausible route to off-target binding and, in principle, mechanism-based side effects. Because both human enzymes are structurally well-characterized, they serve as informative counter-screens for probing PI selectivity.

Here we build an open, reproducible molecular-docking pipeline to estimate the relative binding of a diverse HIV-1 PI library across the viral target and these two human off-targets, quantify selectivity, and ask which molecular properties are associated with it.

2. Materials and Methods

2.1 Target structures

Three crystal structures were obtained from the RCSB Protein Data Bank: HIV-1 protease in complex with ritonavir (PDB **1HXW**; primary target, both chains of the biological homodimer retained), human cathepsin D with pepstatin (PDB **1LYB**; off-target), and human BACE1 with the transition-state inhibitor OM99-2 (PDB **1FKN**; off-target). For each receptor, the co-crystallized inhibitor, crystallographic waters, buffer components, and (for cathepsin D) N-linked glycans were removed, retaining only the biologically relevant protein chains. The geometric centroid of each native inhibitor defined the center of the docking search box, which was padded to at least 22.5 Å per axis.

2.2 Compound library

A library of 20 HIV-1 PIs was assembled from PubChem: 10 FDA-approved agents, one pharmacoenhancer (cobicistat, a ritonavir analog included as a structural comparator), and the remainder discontinued or investigational compounds. Structures were desalted to the largest organic fragment and deduplicated by InChIKey connectivity.

2.3 Structure preparation

Receptors were protonated at pH 7.0 using PDB2PQR with PROPKA-predicted pK_a values, fixing the ionization state of the catalytic aspartate dyad, then converted to the AutoDock PDBQT format with Gasteiger partial charges (Open Babel). Ligands were protonated to their dominant microspecies at pH 7.4, embedded in three dimensions with the RDKit ETKDGV3 method, energy-minimized with the MMFF94 force field, and assigned rotatable-bond torsion trees with Meeko.

2.4 Molecular docking

Docking used AutoDock Vina 1.2.7 (exhaustiveness 8, up to nine poses per run). Each of the 20 ligands was docked against all three receptors ($20 \times 3 = 60$ independent runs), and the most favorable predicted binding energy (ΔG , kcal mol⁻¹) per pair was retained.

2.5 Validation and analysis

The protocol was validated by re-docking each receptor's native inhibitor into its own pocket and measuring heavy-atom root-mean-square deviation (RMSD) from the crystallographic pose. Selectivity was quantified as $\Delta\Delta G = \Delta G_{\text{off-target}} - \Delta G_{\text{HIV}}$; because ΔG is negative, a positive $\Delta\Delta G$ indicates weaker off-target binding and thus greater selectivity for the viral enzyme. The reported selectivity score is the worst-case (smallest) $\Delta\Delta G$ across the two off-targets. Compounds were clustered by their three-receptor binding profile (k-means), and selectivity was correlated (Pearson) against molecular descriptors computed with RDKit.

3. Results

3.1 Protocol validation

Re-docking the native inhibitors reproduced the experimental binding mode for the primary target: ritonavir re-docked into HIV-1 protease with an RMSD of **1.93 Å**, below the conventional 2 Å acceptance threshold, confirming correct box placement and docking parameters. The two off-target native ligands are large, highly flexible peptidic inhibitors (pepstatin and OM99-2); these re-docked less accurately (cathepsin D/pepstatin, 3.69 Å; BACE1/OM99-2 ligand preparation was unsuccessful), a recognized limitation for peptide-like molecules that does not affect docking of the drug-like library compounds.

3.2 Binding-affinity landscape

Across all 60 dockings, predicted affinities ranged from roughly -5.7 to -11.0 kcal mol⁻¹. Mean affinity was strongest for HIV-1 protease (-9.6 kcal mol⁻¹), intermediate for BACE1 (-9.2), and weakest for cathepsin D (-8.8), consistent with the library having been optimized historically against the viral enzyme (Figure 1).

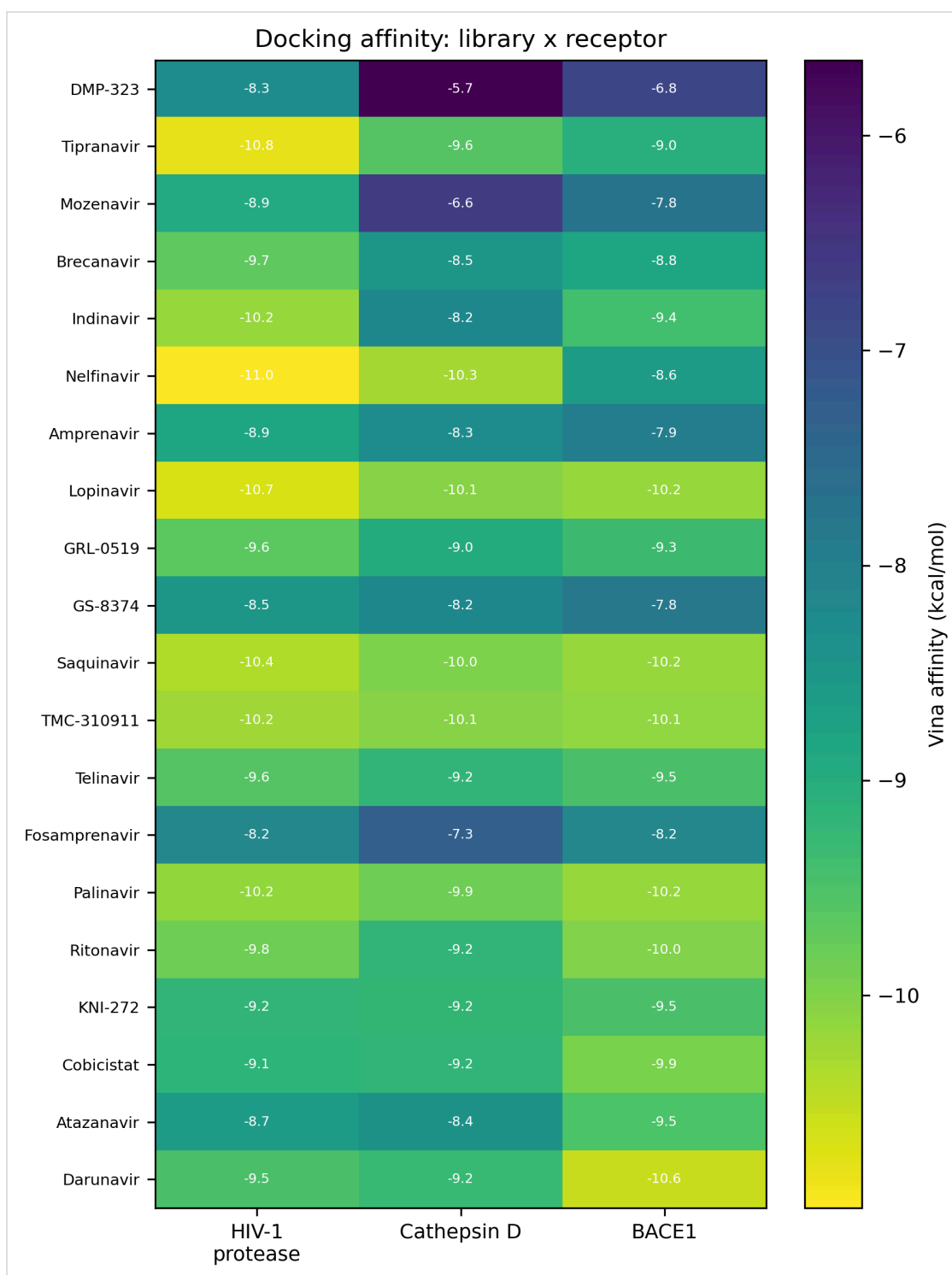


Figure 1. Predicted binding affinity (kcal mol^{-1}) of each compound against the three receptors. Darker cells denote tighter predicted binding. Rows are ordered by selectivity (most HIV-selective at top).

3.3 Selectivity ranking

14 of 20 compounds (70%) were predicted to be HIV-selective, binding the viral protease more tightly than both human off-targets (Table 1, Figure 2). The most selective compounds included the cyclic-urea

inhibitor DMP-323 and the non-peptidic approved drug tipranavir. A minority of compounds — notably darunavir and atazanavir — were predicted to bind an off-target (most often BACE1) at least as tightly as HIV-1 protease; as discussed below, this reflects the limits of docking for absolute discrimination rather than a validated safety liability of these clinically successful drugs.

Table 1. Predicted binding energies and selectivity, ranked most- to least-selective. ΔG in kcal mol⁻¹; selectivity = worst-case $\Delta\Delta G$ (positive = HIV-selective).

Compound	Status	HIV-1 PR	Cathepsin D	BACE1	Selectivity
DMP-323	discontinued	-8.29	-5.65	-6.85	+1.44
Tipranavir	approved	-10.81	-9.58	-9.01	+1.23
Mozenavir	discontinued	-8.91	-6.61	-7.76	+1.15
Brecanavir	discontinued	-9.68	-8.48	-8.79	+0.89
Indinavir	approved	-10.18	-8.19	-9.41	+0.77
Nelfinavir	approved	-10.99	-10.26	-8.60	+0.73
Amprenavir	approved	-8.85	-8.33	-7.92	+0.52
Lopinavir	approved	-10.68	-10.06	-10.17	+0.51
GRL-0519	investigational	-9.64	-8.97	-9.35	+0.29
GS-8374	investigational	-8.52	-8.23	-7.83	+0.29
Saquinavir	approved	-10.37	-9.95	-10.16	+0.21
TMC-310911	investigational	-10.25	-10.06	-10.13	+0.12
Telinavir	discontinued	-9.58	-9.21	-9.49	+0.09
Fosamprenavir	approved	-8.17	-7.28	-8.16	+0.01
Palinavir	discontinued	-10.15	-9.86	-10.16	-0.01
Ritonavir	approved	-9.84	-9.16	-10.00	-0.16
KNI-272	investigational	-9.21	-9.23	-9.50	-0.29
Cobicistat	booster	-9.15	-9.19	-9.94	-0.79
Atazanavir	approved	-8.67	-8.44	-9.52	-0.85
Darunavir	approved	-9.48	-9.25	-10.57	-1.09

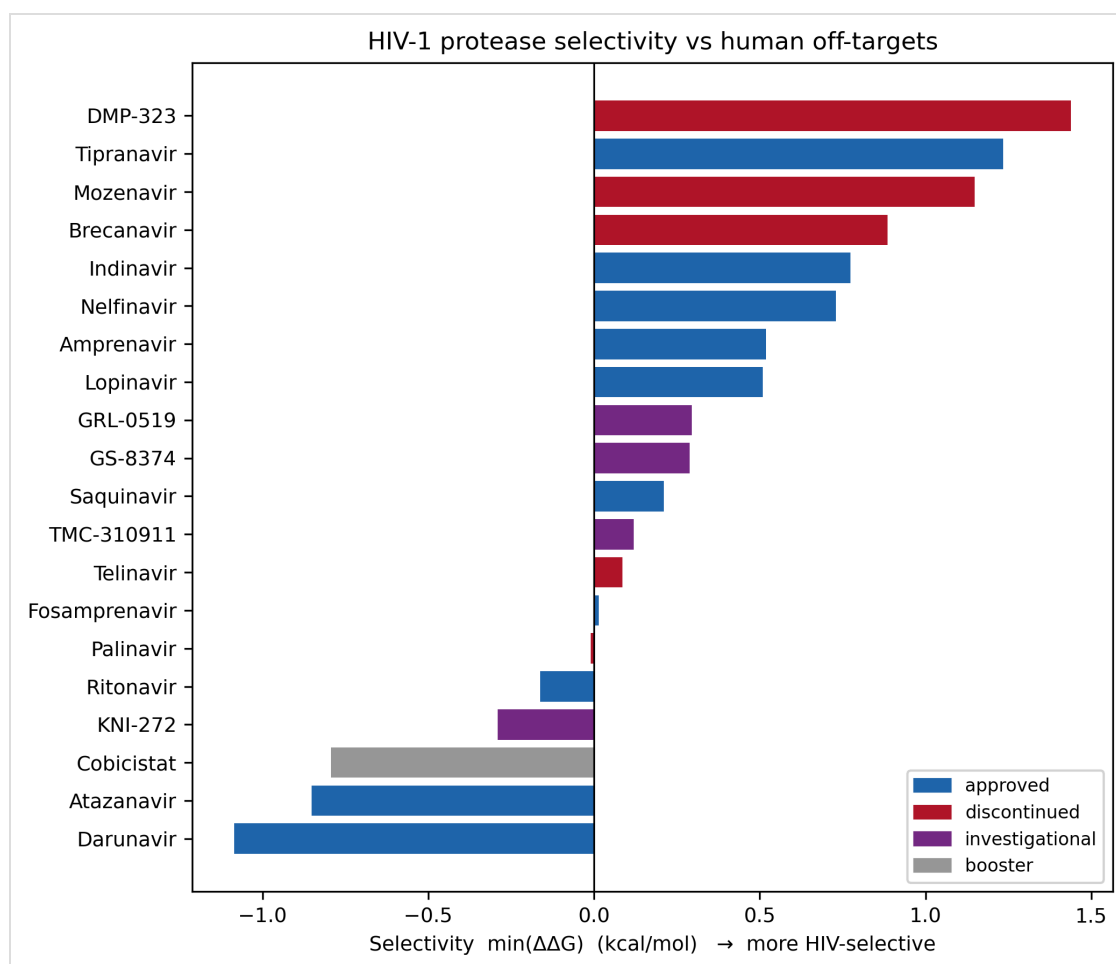


Figure 2. Worst-case selectivity (min $\Delta\Delta G$) per compound, colored by clinical status. Bars to the right indicate preference for HIV-1 protease.

3.4 Structure–selectivity relationships

Predicted selectivity correlated negatively with descriptors of molecular size and polarity (Table 2). The strongest association was with topological polar surface area ($r = -0.59$), followed by rotatable-bond count ($r = -0.49$) and hydrogen-bond acceptors ($r = -0.43$). In other words, smaller, more rigid, and less polar inhibitors tended to prefer the viral pocket, whereas large, flexible, highly polar compounds engaged the human aspartic-protease sites comparably well — a potential design cue for improving selectivity.

Table 2. Pearson correlation of predicted selectivity with molecular descriptors.

Descriptor	Pearson r vs selectivity
Molecular weight	-0.41
logP (lipophilicity)	+0.17
Topological polar surface area	-0.59
H-bond donors	-0.19
H-bond acceptors	-0.43

Rotatable bonds	-0.49
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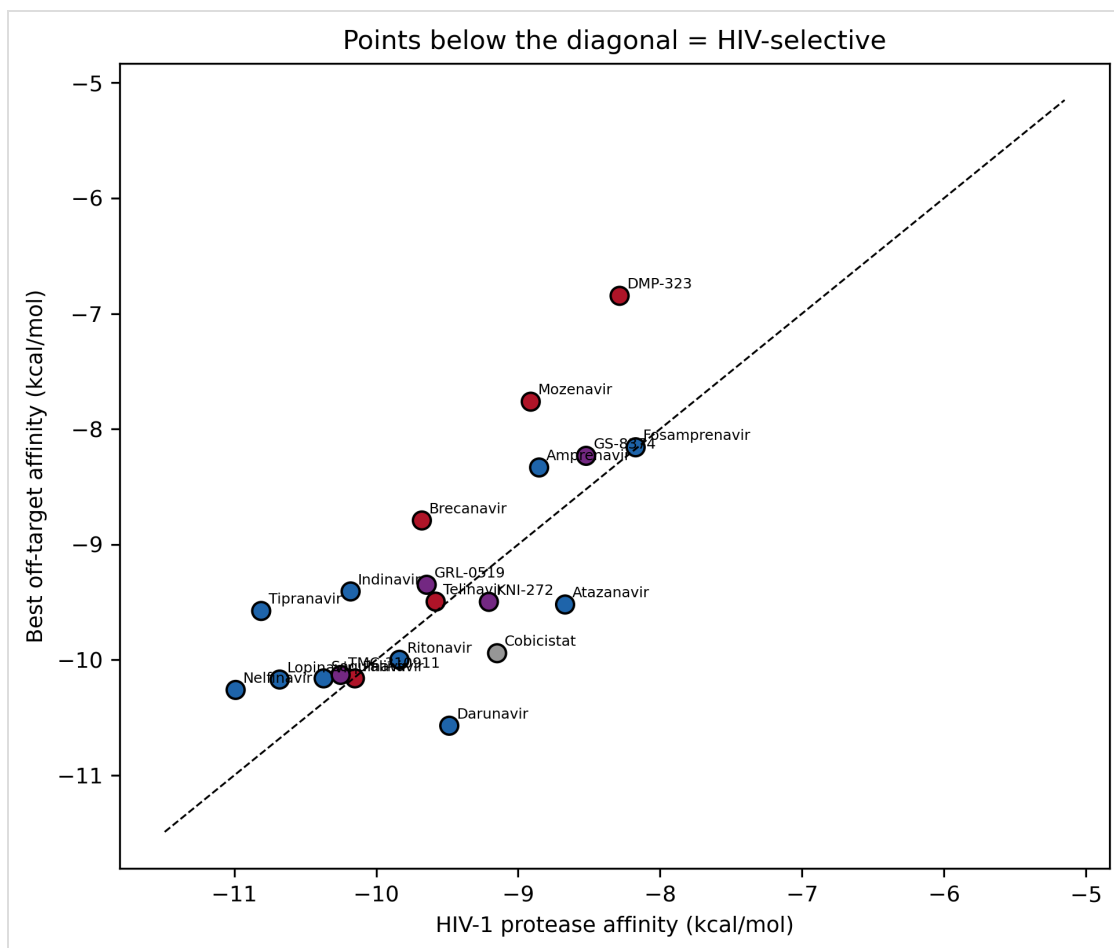


Figure 3. Selectivity landscape: predicted HIV-1 protease affinity versus the best (most favorable) off-target affinity. Points below the diagonal are HIV-selective.

3.5 Binding-profile clustering

Unsupervised k-means clustering of compounds by their three-receptor affinity profile separated the library into groups with distinct cross-reactivity signatures (Figure 4), indicating that selectivity behavior is structured rather than random across the chemotypes represented.

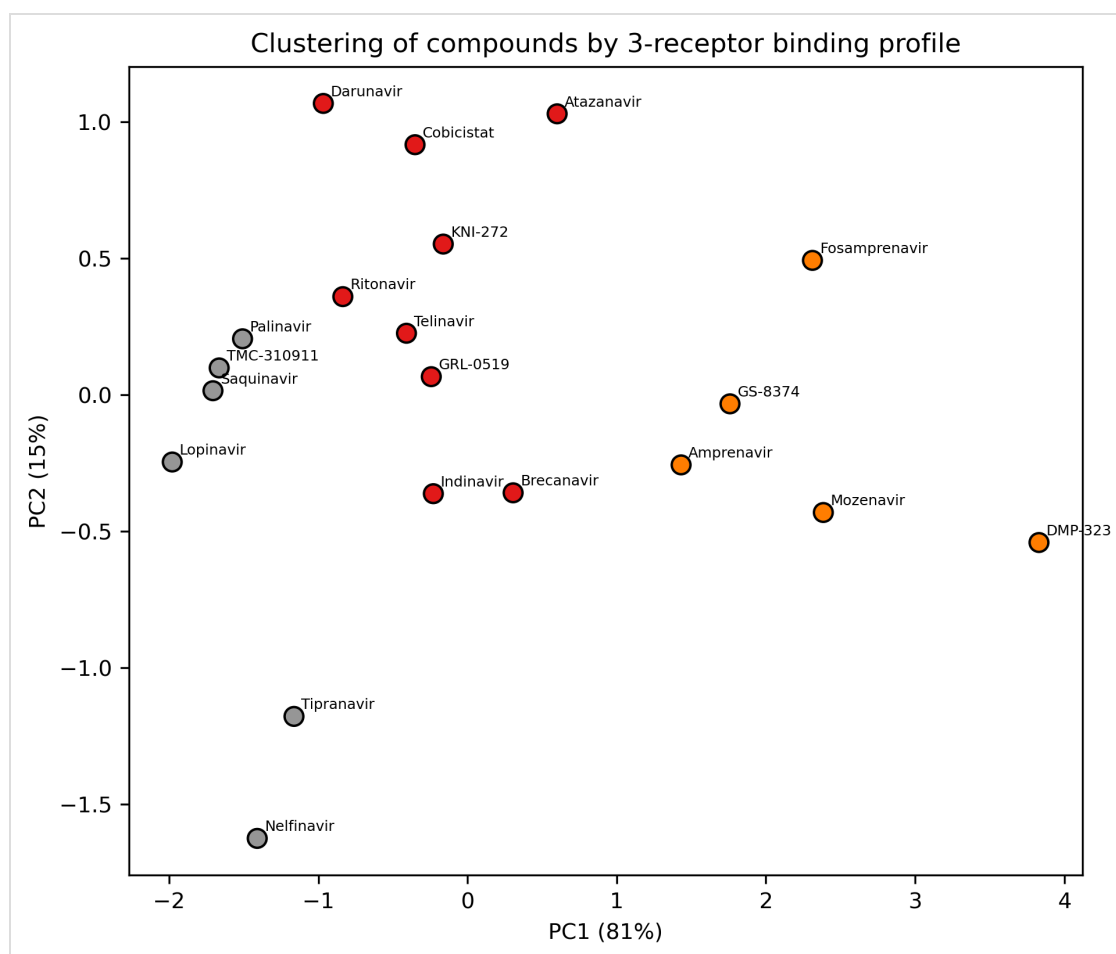


Figure 4. Principal-component projection of the three-receptor binding profiles, colored by k-means cluster.

4. Discussion

The workflow recovered the expected qualitative picture: a library historically optimized against HIV-1 protease binds the viral enzyme most strongly on average, and a majority of compounds are predicted to be selective over the two human aspartic proteases. The structure–selectivity trend — selectivity decreasing with polarity, flexibility, and size — is chemically reasonable: the compact, largely hydrophobic HIV-1 protease pocket may be more discriminating toward small rigid ligands, whereas larger flexible molecules can adopt accommodating conformations in multiple aspartic-protease sites.

Several highly effective clinical drugs (darunavir, atazanavir) were *not* flagged as selective in this purely geometric, single-pose docking analysis. This is an important illustration of docking's limits rather than a contradiction: Vina scores are approximate, neglect protein flexibility and explicit solvation/entropy, and are best used for *relative ranking within* a target rather than for absolute cross-target discrimination. The clinical selectivity of these agents also derives from pharmacokinetics and cellular context not captured by rigid-receptor docking.

5. Limitations

Predicted affinities are docking scores, not experimental free energies, and carry errors of a few kcal mol⁻¹. Receptors were treated as rigid; catalytic-dyad protonation was fixed at a single pH 7 assignment; and a single Vina run (one random seed) was used per pair without replicate averaging. Off-target validation was limited by the difficulty of re-docking large peptidic native ligands. Findings should therefore be read as hypothesis- generating relative trends, not quantitative predictions.

6. Conclusion

Using an entirely open-source pipeline, we mapped the predicted selectivity of 20 HIV-1 protease inhibitors against two human aspartic-protease off-targets, validated the protocol against a crystallographic pose (1.93 Å RMSD), and identified a size/polarity trend associated with viral-target selectivity. The approach provides a transparent, reproducible template for early-stage off-target risk assessment in medicinal chemistry teaching and practice.

Data and code availability

All structures, prepared inputs, docking outputs, analysis scripts, and figures are available at github.com/maxwo123/computational-biology. The pipeline is fully reproducible via the numbered scripts in 04_scripts/.

Software

AutoDock Vina 1.2.7; RDKit 2025.09; Meeko 0.7; PDB2PQR 3.6 with PROPKA 3.5; Open Babel 3.1; PyMOL; pandas; scikit-learn; matplotlib.

References

1. Eberhardt J, Santos-Martins D, Tillack AF, Forli S. AutoDock Vina 1.2.0. *J Chem Inf Model* 2021;61:3891–3898.
2. Trott O, Olson AJ. AutoDock Vina. *J Comput Chem* 2010;31:455–461.
3. Kempf DJ, et al. ABT-538 (ritonavir) and HIV protease. *Proc Natl Acad Sci USA* 1995;92:2484–2488. (PDB 1HXW)
4. Baldwin ET, et al. Crystal structure of human cathepsin D. *Proc Natl Acad Sci USA* 1993;90:6796–6800. (PDB 1LYB)
5. Hong L, et al. Structure of BACE1–OM99-2 complex. *Science* 2000;290:150–153. (PDB 1FKN)
6. Dolinsky TJ, et al. PDB2PQR. *Nucleic Acids Res* 2007;35:W522–W525.
7. Olsson MHM, et al. PROPKA3. *J Chem Theory Comput* 2011;7:525–537.
8. O'Boyle NM, et al. Open Babel. *J Cheminform* 2011;3:33.
9. Forli S, et al. Computational protein–ligand docking with AutoDock/Meeko. *Nat Protoc* 2016;11:905–919.
10. Landrum G, et al. RDKit: Open-source cheminformatics. rdkit.org.
11. Ghosh AK, et al. Darunavir and the design of HIV-1 protease inhibitors. *Acc Chem Res* 2008;41:78–86.
12. Wlodawer A, Vondrasek J. Inhibitors of HIV-1 protease. *Annu Rev Biophys Biomol Struct* 1998;27:249–284.